ERIK KOVÁCS

STUDENT

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github.com/KovacsErik-Gh

4th-year Automation and Computer Science student with hands-on experience in deep learning, computer vision, and autonomous systems, complemented by industrial experience building internal business applications at Tenaris Silcotub. Quick to pick up new tools and technologies, and highly motivated to grow within a professional environment, learn from experienced engineers, and contribute to real-world projects in a collaborative team.

Experience

App Developer Intern Tenaris Silcotub, Zalău July 13, 2026 – September 18, 2026 (10 weeks)

- Built internal business applications in Microsoft Power Apps, translating end-user requirements into working solutions
- Automated workflows and approval processes using Power Automate
- Managed and integrated company data and documents in SharePoint
- Completed a project management training, applying its principles to planning and delivering my tasks

Personal & Academic Projects

GymTech – Gym Management System (C#, ASP.NET Core 8, EF Core, Angular, SQL Server) GitHub

- Contributed, in a team, to a full-stack gym management platform (ASP.NET Core Web API, Angular frontend) with role-based access for Admin, Receptionist, Trainer and Member
- Platform covers subscriptions, payments, attendance check-ins and trainer assignment, backed by xUnit, Vitest and Playwright test suites

Guess the Flag – Web Quiz Application (Python, FastAPI, Node.js, Express, SQLite) GitHub

- Built a full-stack flag-guessing quiz (195 countries) with a decoupled architecture: FastAPI backend and Express/EJS frontend
- Implemented JWT authentication with bcrypt password hashing and a SQLite-backed global leaderboard with per-user score history

Adaptive Cruise Control with Computer Vision (Python, PyTorch, YOLOv8, OpenCV) GitHub

- Developed an autonomous navigation system combining semantic segmentation (FCN-ResNet50) and object detection (YOLOv8) for real-time scene understanding, achieving 90.71% pixel-level accuracy across 12 classes
- Implemented adaptive speed control with safety-first hierarchical decision logic (emergency stop, braking, weather adaptation), optimized for CPU-only execution and trained on the CARLA dataset

DC Motor PID Controller Simulation (MATLAB/Simulink)

- Modeled DC motor dynamics and performed system identification, then designed a PID controller tuned with the Ziegler-Nichols method
- Achieved 2% overshoot and 0.5 s settling time in simulation

Education

Technical University of Cluj-Napoca, Faculty of Automation and Computer Science – BSc 2023 – 2027 (expected)

Silvania National College, Zalău (Baccalaureate Diploma)

Training & Certifications

Project Management Training – Ascendis, Certificate of Completion 2026

Technical Skills

Programming Languages: Python, C#, Java, C/C++, SQL, TypeScript, MATLAB

Frameworks & Libraries: ASP.NET Core, Entity Framework Core, Angular, FastAPI, Node.js/Express, xUnit, PyTorch, OpenCV, YOLOv8, NumPy, Pandas

Low-Code / Microsoft 365: Power Apps, Power Automate, SharePoint

Tools & Platforms: Git/GitHub, Jupyter Notebook, Anaconda, AutoCAD, Simulink

Core Concepts: Deep Learning, Computer Vision (Semantic Segmentation, Object Detection), Control Systems, Process Automation, Project Management

Soft Skills

Analytical Thinking • Teamwork • Communication • Problem Solving • Time Management • Adaptability • Eagerness to Learn

Languages

Hungarian (Native) • Romanian (C1) • English (C1) • German (A2)